Chapter04

Image Representation in Computer System

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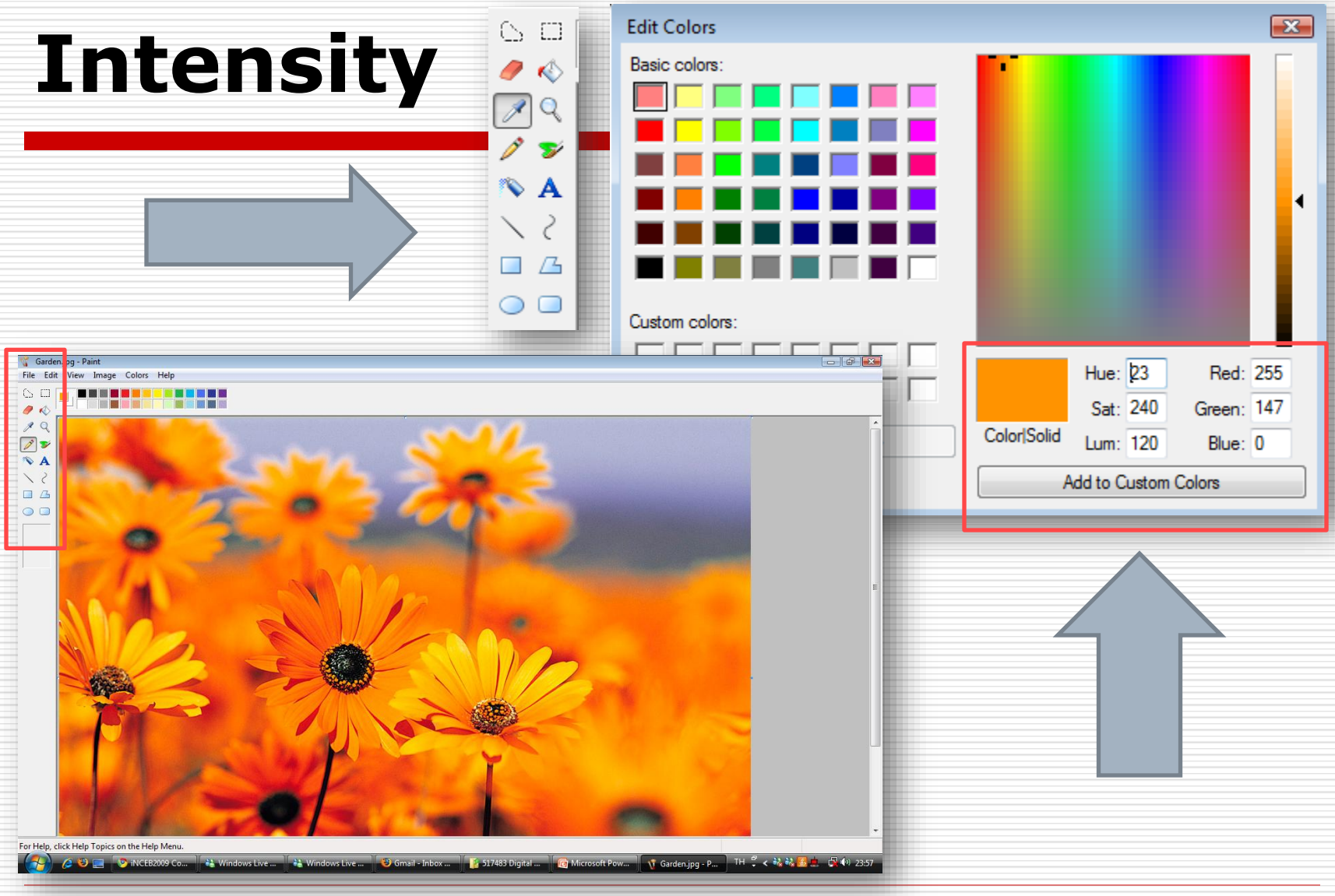
Topics

- Image Terminology.
- Picture File Format.
- RGB Color Image or True Color Image.
- Gray Scale Image.
- Black and White Image.
- 4-16-256-Indexed Image.
- Conclusion.

Image Terminology

- **Pixel** – Picture Cell, Picture Element.
- **Intensity** – Color Value in each pixel of Image.
- **Resolution** – Image Resolution, Display Screen Resolution.
- **RGB** – Red Green Blue.
- **Background**.
- **Foreground**.

Intensity


Edit Colors
Basic colors:
Custom colors:
Color|Solid
Hue: 23
Sat: 240
Lum: 120
Red: 255
Green: 147
Blue: 0
Add to Custom Colors
Garden.jpg - Paint
File Edit View Image Colors Help
For Help, click Help Topics on the Help Menu.

Picture File Formats

- **Bitmap Image (*.BMP)**
  - Microsoft Standard Image File Format.
  - Supported on MS-Windows and MS-Application.
  - Unsupported on Mac.
- **Picture Image (*.PICT)**
  - Apple Standard Image File Format.
  - Supported on Mac OS and Mac Application.
  - Unsupported on PC.

Picture File Formats (Con't)

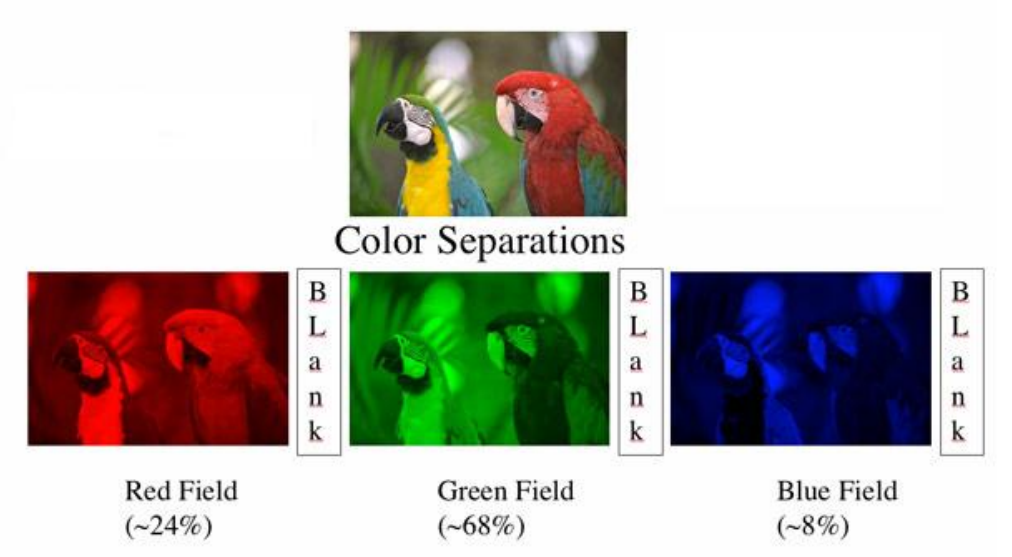
- **Tagged Image File Format (*.TIFF)**
  - Bitmap Image File Format on 1986.
  - Supported both Microsoft and Apple.
  - Popular in PC and Mac.
- **Joint Photographic Expert Group (*.JPEG)**
  - Used on Web Browser, Monitor and Digital Camera.
  - Image Compression.
  - Shorted name is JPG.

Picture File Formats (Con't)

- **Graphic Interchange Format (*.GIF)**
  - Developed by CompuServe
  - Used on Computer Animation and Video Frame.
- **Portable Network Graphic (*.PNG)**
  - Vector File Format.
  - Alternative GIF Image.
  - Image sized is lower than JPEG.
  - Flexible for Image Resizable, Zoom in, Zoom out in High Level Percentages.

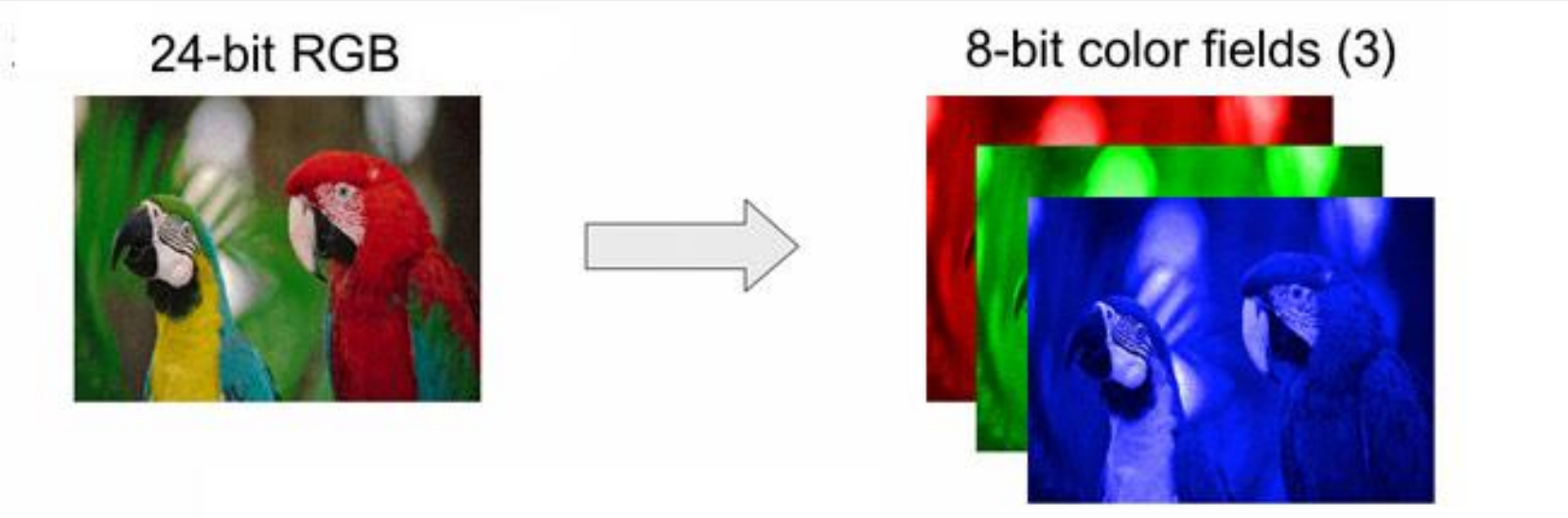
In This Class

- In this class, we will introduce image processing for only BMP and JPEG.
- **JPEG** → Chapter of Image Compression.



RGB Color Planes

- One picture composed by 3-Planes Color Vector → RGB.
- R is Red Color, G is Green Color and B is Blue Color.



RGB Color Values

- One pixel kept Color Values in RGB Term.
- In each color has 256 color values.
- 256 Color Values can be written in 2 Mode.
  - 0 – 255 in Decimal based number.
  - 00 – FF in Hex based number.
- Maximum Colors of RGB Model.
  - 256 x 256 x 256 color values.
  - 16.7 million color values.

RGB Color Values (Con't)

Color	Color Value (Base10)	Color Value (Base16)	Memory (Byte)
Red	0 – 255	00 – FF	1
Green	0 – 255	00 – FF	1
Blue	0 – 255	00 – FF	1

RGB Color Values (Con't)

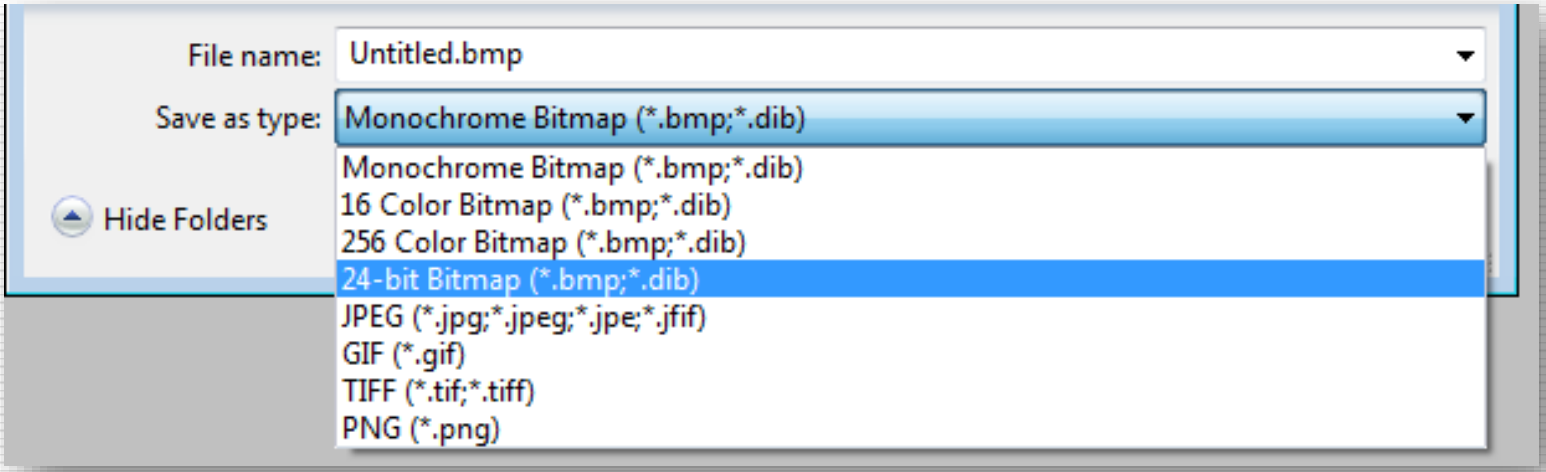
- **Black Color**
  - 10-Based → (R,G,B) → (0,0,0)
  - 16-Based → RGB → 000000
- **White Color**
  - 10-Based → (R,G,B) → (255,255,255)
  - 16-Based → RGB → FFFFFF

RGB Color Values (Con't)

RGB Value			Actual Color
Red	Green	Blue	
0	0	0	Black
255	255	255	White
255	255	0	Yellow
255	130	255	Pink
146	81	0	Brown
157	95	82	Purple
140	0	0	Maroon

Image Representation

- RGB Image or True Color Image.
- Gray Scale Image.
- Black and White Image (B/W).
- Indexed Color Image



Color Image Processing

- **Image Normalization.**
  - RGB Image → Gray Scaled Image
- **Image Binarization.**
  - Gray Scaled Image → Black/White Image

Color Image or RGB Image

Color image or RGB image
each pixel contains a vector representing red, green and blue components.

RGB components

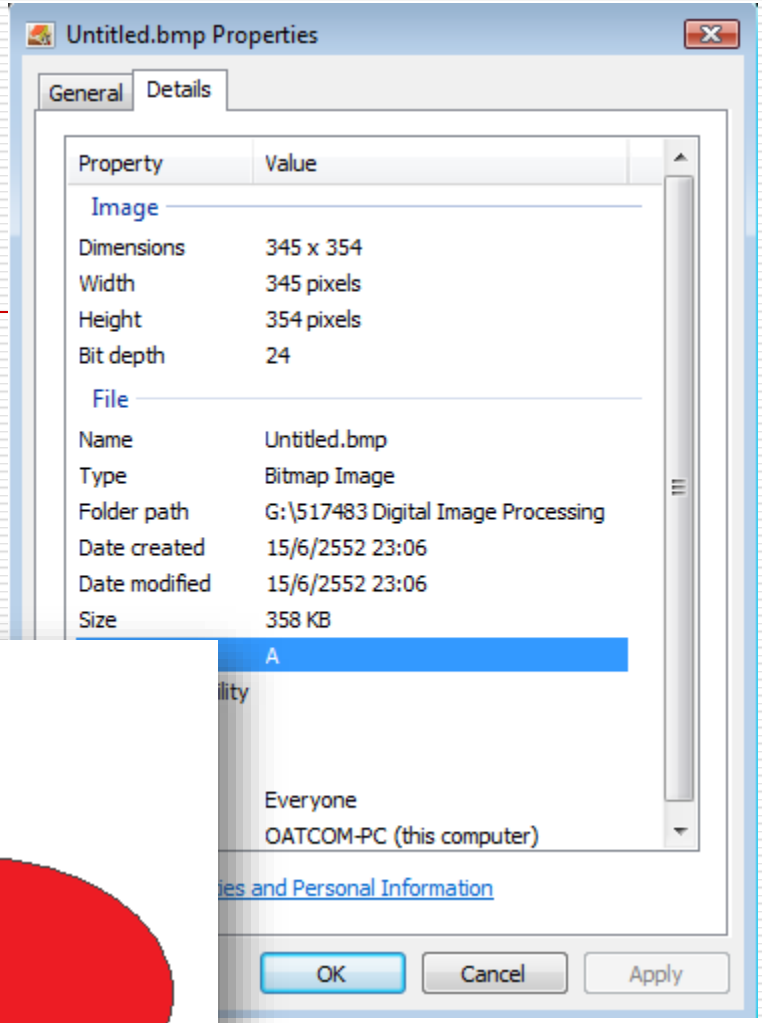
$$\begin{bmatrix} 10 & 10 & 16 & 28 \\ 9 & & & \\ 15 & & & \\ 32 & & & \end{bmatrix} \begin{bmatrix} 65 & 70 & 56 & 43 \\ 32 & & & \\ 21 & & & \\ 54 & & & \end{bmatrix} \begin{bmatrix} 99 & 70 & 56 & 78 \\ 60 & 90 & 96 & 67 \\ 85 & 85 & 43 & 92 \\ 32 & 65 & 87 & 99 \end{bmatrix}$$

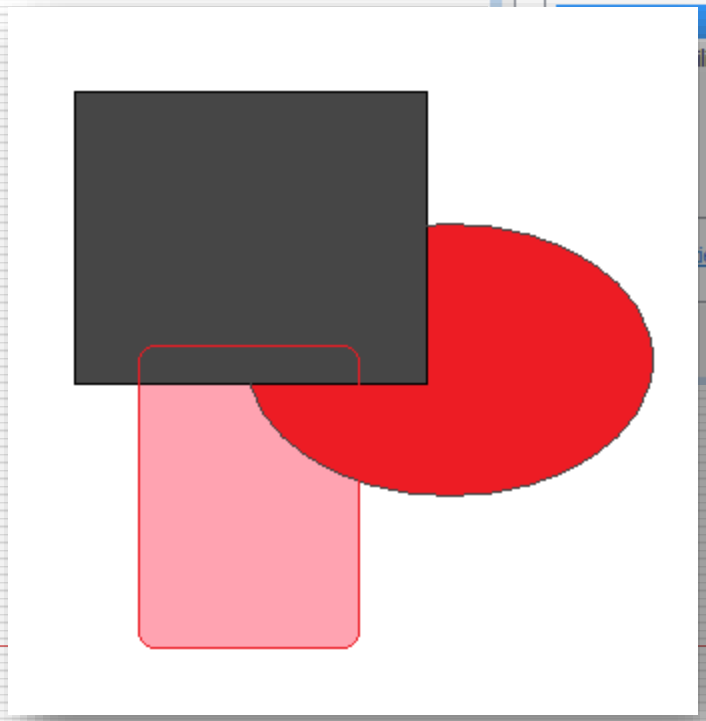
RGB Image File Sized

- **1 pixel**
  - 1 pixel kept RGB Color values.
  - R = 8 bits, G = 8 bits, B = 8 bits.
  - RGB = 3 x 8 = 24 bits = 3 bytes.
  - File size / 1 pixel = 3 bytes.
- **n x m pixel**
  - File size = n x m x 3 = 3nm bytes.

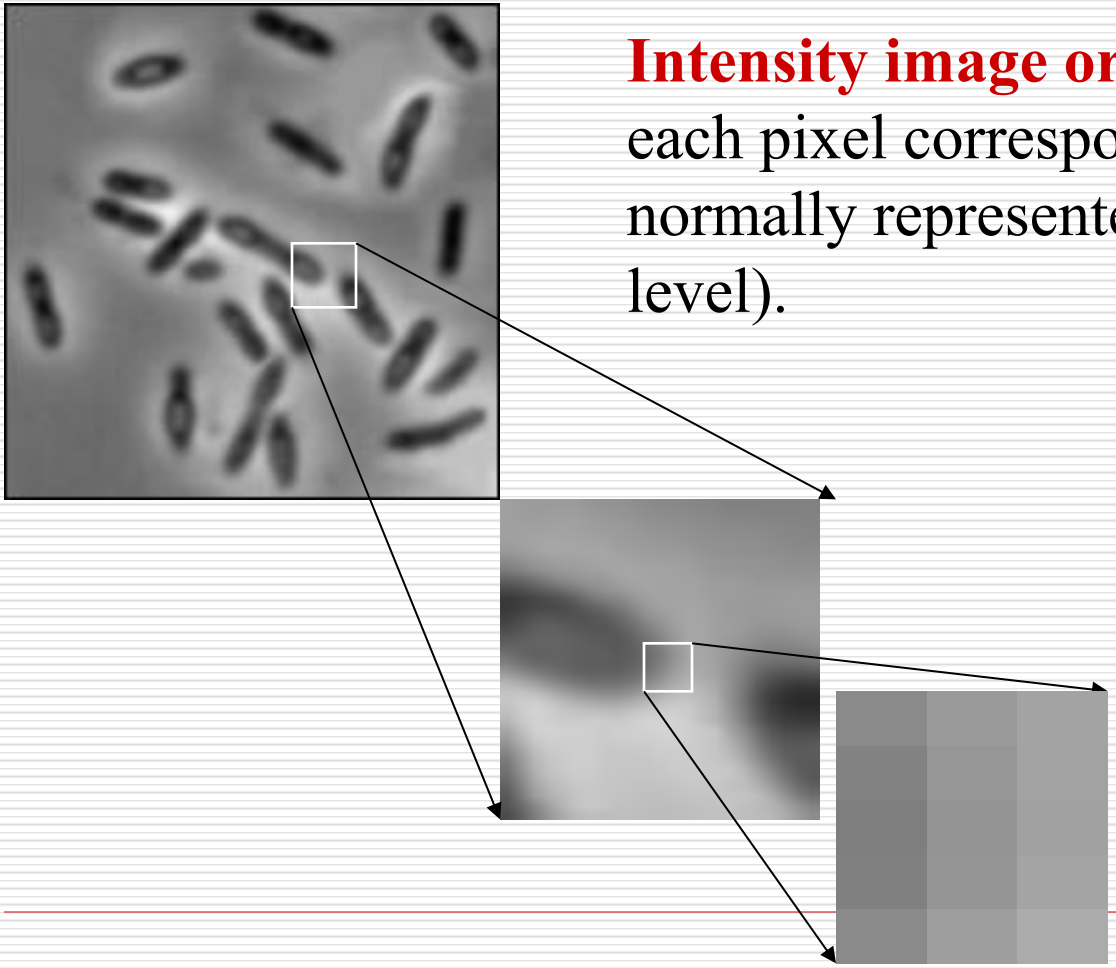
Bitmap Size

- 3 x 345 x 354
- = 366390 bytes.
- = 358 KB





Gray Scale Image



Intensity image or monochrome image
each pixel corresponds to light intensity normally represented in gray scale (gray level).

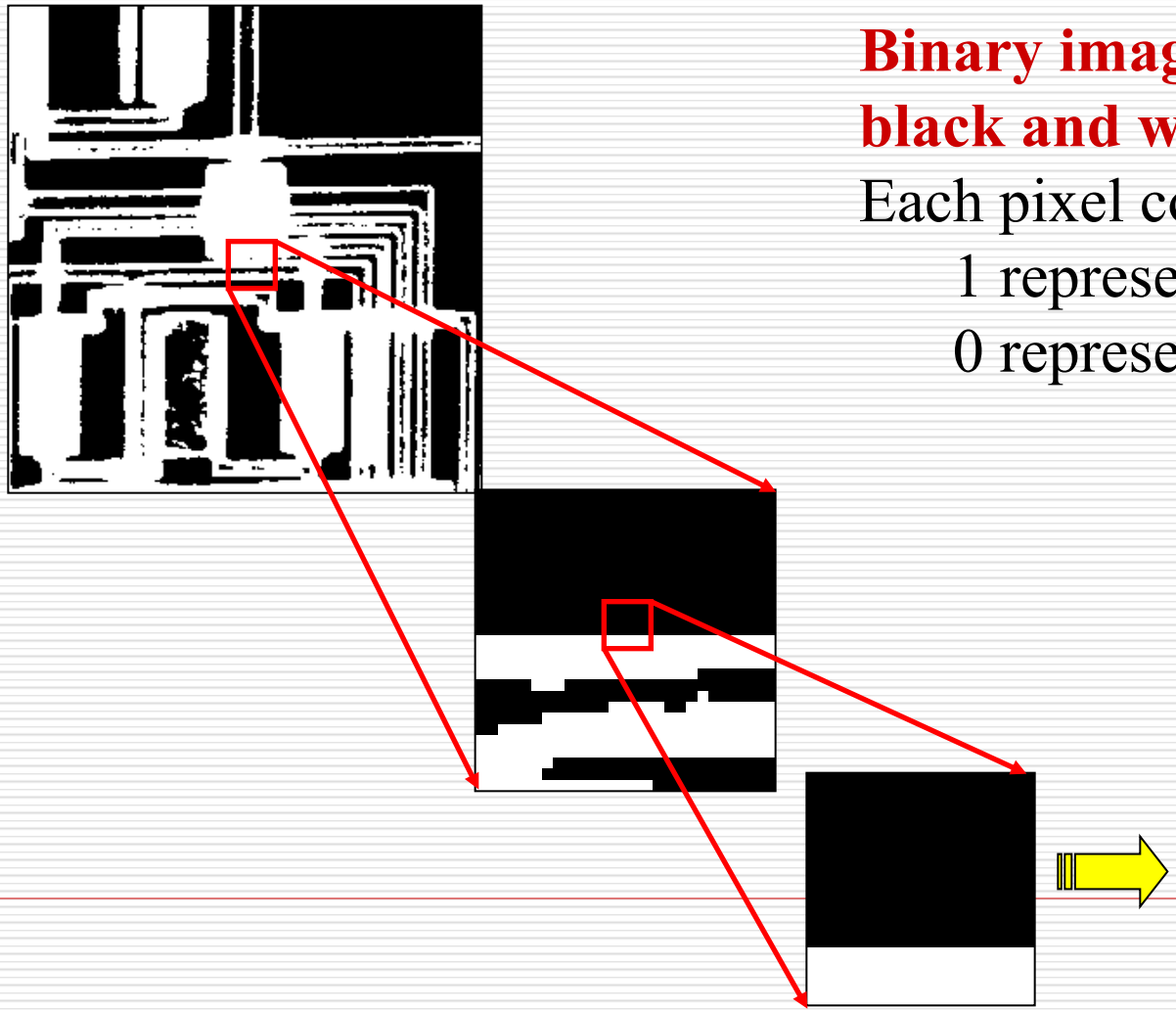
Gray scale values

$$\begin{bmatrix} 10 & 10 & 16 & 28 \\ 9 & 6 & 26 & 37 \\ 15 & 25 & 13 & 22 \\ 32 & 15 & 87 & 39 \end{bmatrix}$$

Gray Scale Image File Sized

- **1 pixel**
  - 1 pixel kept Gray Scale Color values.
  - Range(GrayScaleValue) = 0 – 255.
  - 0 – 255 is used 8 bits representation.
  - Gray Scale = 1 x 8 = 8 bits = 1 byte.
  - File size / 1 pixel = 1 byte.
- **n x m pixel**
  - File size = n x m x 1 = nm bytes.

Black and White Image



Binary image or
black and white image

Each pixel contains one bit :

1 represent white

0 represents black

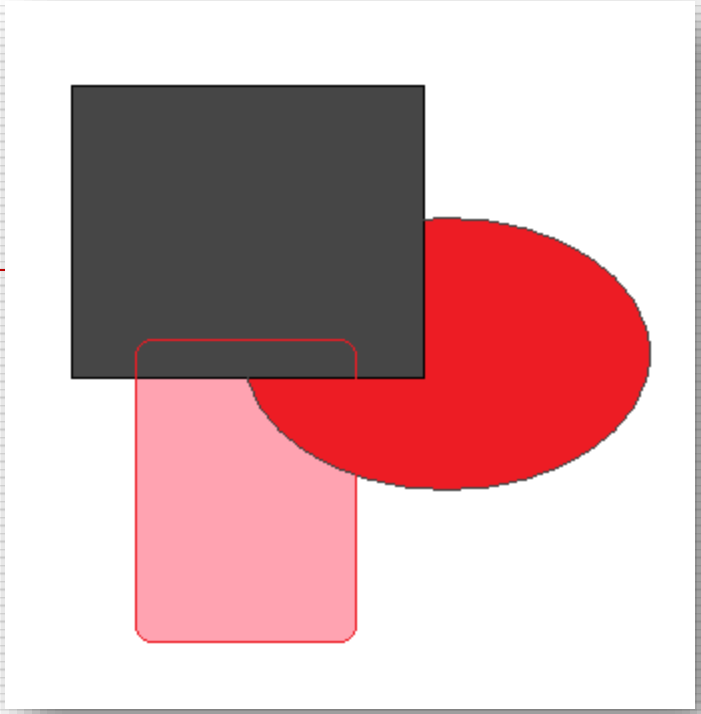
Binary data

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

B/W Image File Sized

- **1 pixel**
  - 1 pixel kept Black or White.
  - Black is 0, White is 1.
  - 0 - 1 is used 1 bit representation.
  - B/W = 1 bit = 1/8 byte.
  - File size / 1 pixel = 1/8 byte.
- **n x m pixel**
  - File size = n x m x 1/8 = nm/8 bytes.

RGB Problem???



- This image sized is
  - 3 x 345 x 354
  - = 366390 bytes.
  - = 358 KB
- This image has 4 color
  - White, Black, Red and Pink
  - = ¼(345 x 354) + (4 x 3)
  - = 29 KB

4 Indexed Color Image

- Converted RGB -> 4 Indexed Color
- Create color table of 4 Indexed

#Index	Color Name	RGB
0	Black	(0,0,0)
1	Red	(255,0,0)
2	Pink	(255,130,255)
3	White	(255,255,255)

- Table sized is 4 x 3 = 12 bytes.

4 Indexed Color Image

- In each pixel kept
  - 0,1,2,3 in Decimal Number or
  - 00,01,10,11 in Binary Number (2 bits).
- 1 pixel of 4 Indexed Color
  - used 2 bits/pixel = 2/8 = ¼ bytes.
- 4 Indexed Color
  - nm pixel, file sized is image sized + table color sized = ¼nm + (4x3).

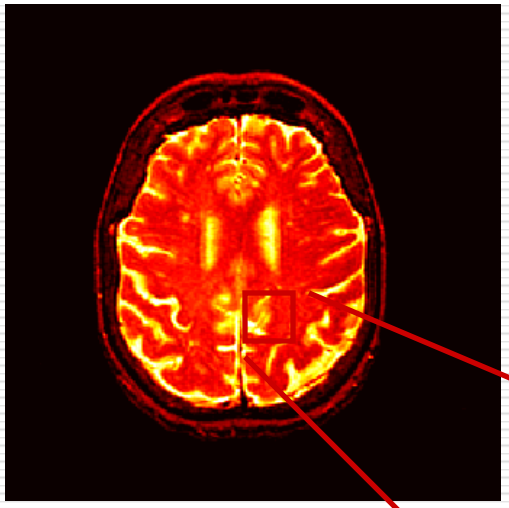
16 Indexed Color Image

- In each pixel kept
  - 0,1,2,3,…,15 in Decimal Number or
  - 0000,0001,0010,…,1111 in Binary Number (4 bits).
- 1 pixel of 16 Indexed Color
  - used 4 bits/pixel = 4/8 = ½ bytes.
- 16 Indexed Color
  - nm pixel, file sized is image sized + table color sized = ½nm + (16x3).

256 Indexed Color Image

- In each pixel kept
  - 0,1,2,3,…,255 in Decimal Number or
  - 00000000,00000001,00000010,…, 11111111 in Binary Number (8 bits).
- 1 pixel of 256 Indexed Color
  - used 8 bits/pixel = 8/8 = 1 bytes.
- 256 Indexed Color
  - nm pixel, file sized is image sized + table color sized = nm + (256x3).

Indexed Color Image



Index image

Each pixel contains index number pointing to a color in a color table

$$\begin{bmatrix} 1 & 4 & 9 \\ 6 & 4 & 7 \\ 6 & 5 & 2 \end{bmatrix}$$

Index value

Index No.	Red component	Green component	Blue component
1	0.1	0.5	0.3
2	1.0	0.0	0.0
3	0.0	1.0	0.0
4	0.5	0.5	0.5
5	0.2	0.8	0.9
...	...	...	...

Color Table

Total of Image Sized

Color Image Type	Image Memory (Byte)	Table Memory (Byte)	Total Memory (Bytes)
RGB Image Color	n x m x 3	0	n x m x 3
Gray Scale Image Color	n x m x 1	0	n x m x 1
Black & White Image Color	n x m x 1/8	0	n x m x 1/8
4-Indexed Image Color	n x m x 1/4	4 x 3	(nxm/4) + (4x3)
16-Indexed Image Color	n x m x 1/2	16 x 3	(nxm/2) + (16x3)
256-Indexed Image Color	n x m x 1	256 x 3	(nxmx1) + (256x3)

Example I

- ภาพจะต้องมีขนาดอย่างต่ำเป็นเท่าไหร่ จึงจะทำให้การเก็บภาพแบบ 16 สี คุ้มกว่าการเก็บภาพแบบ True Color Image
- หมายเหตุ ภาพนั้นมีสีเพียง 16 ระดับเท่านั้น

$$
\begin{aligned}
(3x16) + 0.5nm &< 3nm \\
3nm - 0.5nm &> 48 \\
2.5nm &> 48 \\
nm &> 48/2.5 \\
nm &> 19.2 \qquad \textbf{\underline{Ans}}
\end{aligned}
$$

- เพราะฉะนั้นภาพต้องมีขนาดอย่างต่ำ 20 pixel ขึ้นไป

Example II

- ภาพจะต้องมีขนาดอย่างต่ำเป็นเท่าไหร่ จึงจะทำให้การเก็บภาพแบบ 256 สี คุ้มกว่าการเก็บภาพแบบ True Color Image
- หมายเหตุ ภาพนั้นมีสีเพียง 256 ระดับเท่านั้น

$$
\begin{aligned}
(3x256) + nm &< 3nm \\
3nm - nm &> 768 \\
2nm &> 768 \\
nm &> 768/2 \\
nm &> 384 \qquad \textbf{\underline{Ans}}
\end{aligned}
$$

- เพราะฉะนั้นภาพต้องมีขนาดอย่างต่ำ 385 pixel ขึ้นไป

Example III

- ภาพจะต้องมีขนาดอย่างต่ำเป็นเท่าไหร่ จึงจะทำให้การเก็บภาพแบบ 2 สี คุ้มกว่าการเก็บภาพแบบ B/W Image
- หมายเหตุ ภาพนั้นมีสีเพียง 2 ระดับสี และเป็นสีดำกับขาวเท่านั้น

  $(2x3) + nm/8 \quad < nm/8$
- ไม่สามารถแก้อสมการได้ ดังนั้นจึงไม่มีคำตอบของอสมการนี้
- หรือพูดง่ายๆ คือ ไม่มีทางเป็นไปได้ในการที่เราจะเก็บภาพแบบ 2 รหัสสี แล้วคุ้มกว่าการเก็บแบบ B/W เนื่องจากมีแค่ค่าสี 2 ระดับ คือ ขาวดำ เก็บแบบ B/W ใช้น้อยกว่าอยู่แล้ว หากเราไปจัดเก็บแบบ 2 รหัสสี จะต้องมีการเสียพื้นที่ของตารางสีในการจัดเก็บรวมอยู่ด้วย